

# Lecture 03b – One-way ANOVA

ENVX2001 Applied Statistical Methods

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## From $t$ -tests to ANOVA

## From two groups to four

Before the break we compared two cattle breeds. But what if there were four breeds instead of two?

- Can we still use  $t$ -tests? **Yes**, but we would need to test every pair
- With 4 groups that is  $\binom{4}{2} = 6$  comparisons: 1 vs 2, 1 vs 3, 1 vs 4, 2 vs 3, 2 vs 4, 3 vs 4

The problem: every test carries a risk of a **Type I error** (a false positive).

- We set  $\alpha = 0.05$ , so each comparison has a 5% chance of rejecting  $H_0$  when it is actually true
- Six tests means six chances to get it wrong

## The multiple comparisons problem

With 6 groups there are  $\binom{6}{2} = 15$  pairwise comparisons. The probability that **none** produce a false positive is  $0.95^{15}$ , so:

$$P(\text{at least one false positive}) = 1 - 0.95^{15} = 53.7\%$$

| Groups | Pairwise tests | P(at least one false positive) |
|--------|----------------|--------------------------------|
| 2      | 1              | 5.0%                           |
| 4      | 6              | 26.5%                          |
| 6      | 15             | 53.7%                          |
| 8      | 28             | 76.2%                          |
| 10     | 45             | 90.1%                          |

- With 10 groups, a false positive is almost guaranteed
- We need a method that tests all groups at once

# Chick feeding

## The experiment



We need a method that handles more than two groups. Consider a new experiment:

- 20 chicks are randomly assigned to one of 4 diets
- 5 replicates per diet
- Response: weight gain (g)

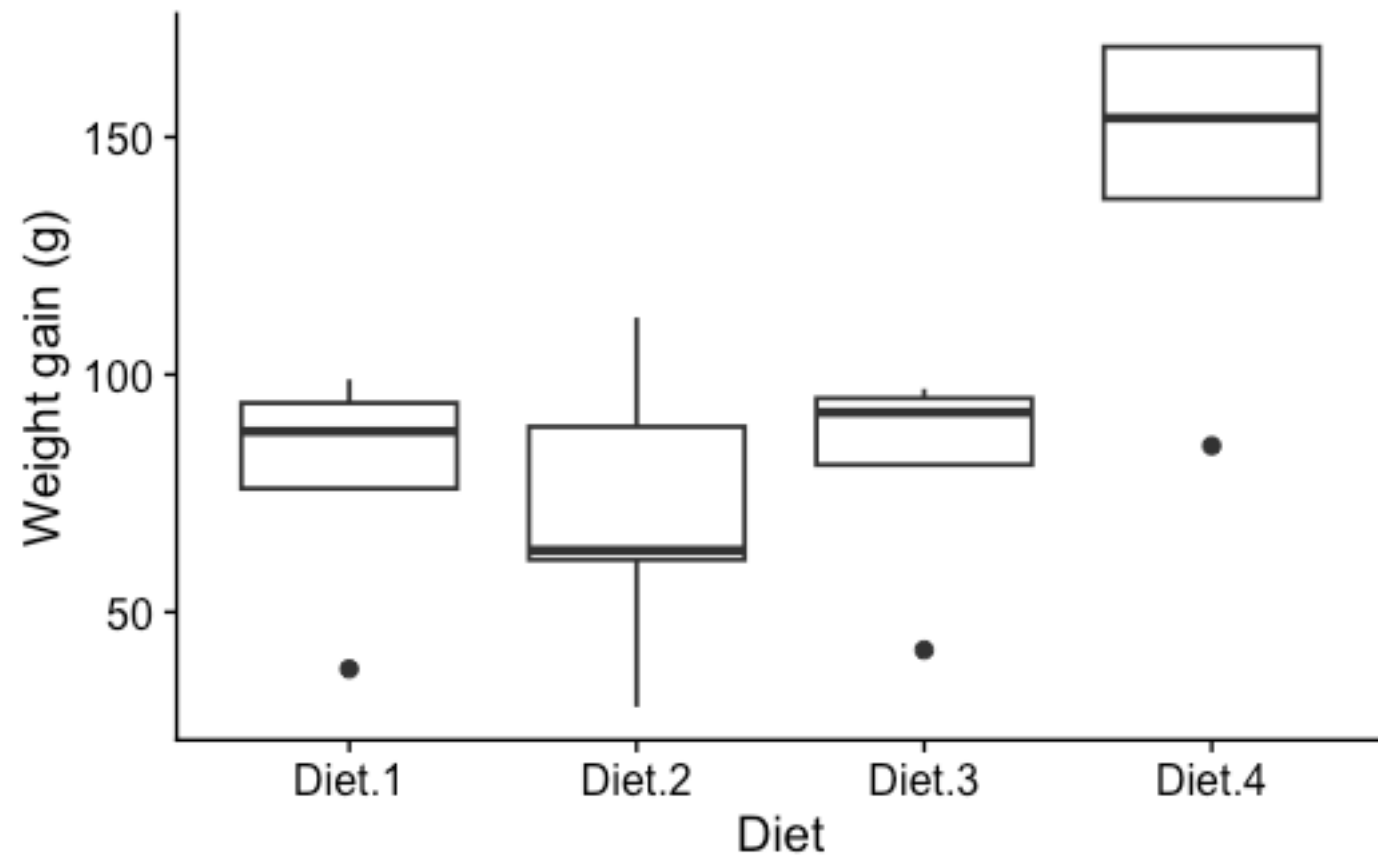
```
chicks <- read.csv("data/chicks.csv") |>  
  pivot_longer(  
    starts_with("Diet"), names_to = "diet", values_to = "weight"
```

```
) |>  
mutate(diet = as.factor(diet))
```

## Are there differences?

```
ggplot(chicks, aes(diet, weight)) +  
  geom_boxplot() +  
  labs(x = "Diet", y = "Weight gain (g)") +  
  cowplot::theme_cowplot()
```





What do the boxplots suggest?

## Terminology

| Term              | Meaning                           | In this experiment             |
|-------------------|-----------------------------------|--------------------------------|
| <b>Factor</b>     | Categorical variable being tested | Diet                           |
| <b>Levels</b>     | Categories within a factor        | Diet 1, Diet 2, Diet 3, Diet 4 |
| <b>Replicates</b> | Observations per level            | $r = 5$                        |

This is a **one-way** ANOVA because there is only one factor. **What is ANOVA?**

# **HATPC for Analysis of Variance**

## H: Hypotheses

In the  $t$ -test we asked whether **two** means were equal. ANOVA extends this to **any number** of groups.

- **Null hypothesis:**  $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$
- **Alternative hypothesis:**  $H_1$  : not all  $\mu_i$  are equal

The alternative does not say all means differ. It says at least two do.

**Model equation:**

$$y_{ij} = \mu_i + \varepsilon_{ij}$$

where  $i = 1, \dots, t$  treatments and  $j = 1, \dots, n_i$  replicates. The same structure as the  $t$ -test model, but  $i$  now ranges over more than two groups.

## A: Assumptions

The same three assumptions from the  $t$ -test apply here. We fit the model first, then check.

```
model ← aov(weight ~ diet, data = chicks)
```

**Normality** (Shapiro-Wilk on residuals):

```
shapiro.test(residuals(model))
```

Shapiro-Wilk normality test

```
data: residuals(model)
W = 0.90961, p-value = 0.06265
```

$p > 0.05$ : no evidence against normality.

**Equal variances** (Bartlett's test):

```
bartlett.test(weight ~ diet, data = chicks)
```

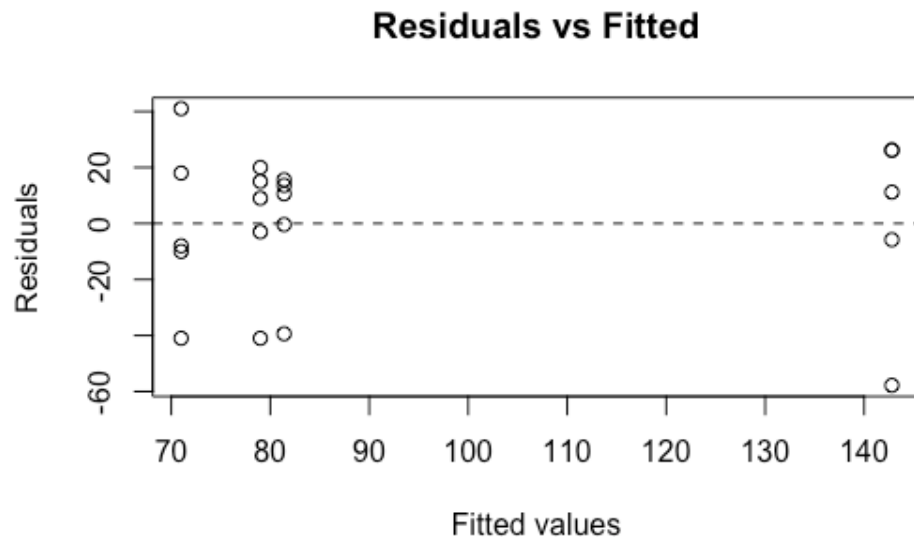
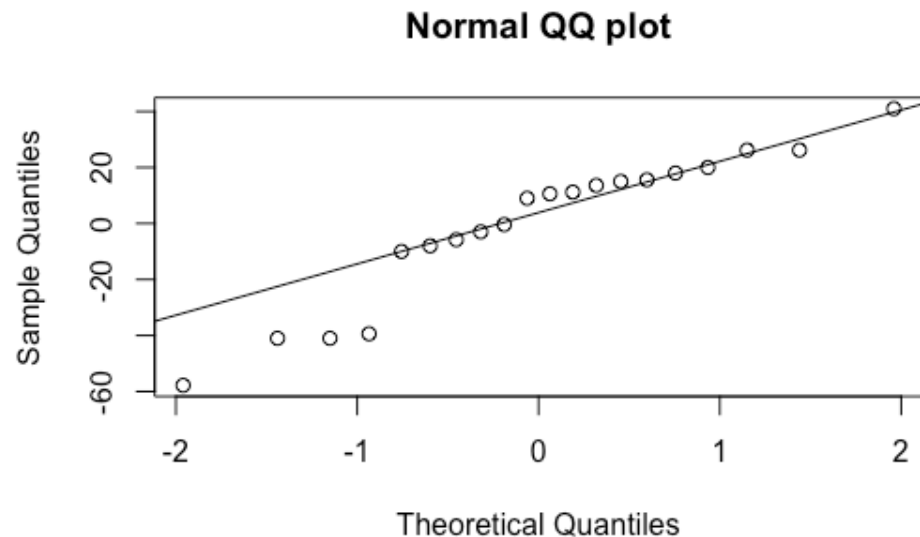
Bartlett test of homogeneity of variances

```
data: weight by diet
Bartlett's K-squared = 0.85164, df = 3, p-value
= 0.8371
```

$p > 0.05$ : variances are similar across groups.

## A: Residual diagnostics

```
qqnorm(residuals(model), main = "Normal QQ plot")
qqline(residuals(model))
plot(fitted(model), residuals(model),
     xlab = "Fitted values", ylab = "Residuals",
     main = "Residuals vs Fitted")
abline(h = 0, lty = 2)
```



- QQ plot: points follow the line, supporting normality
- Residuals vs fitted: no fan shape, supporting equal variances
- Independence holds by design: each chick is measured once

## A: A shortcut for next time

R can produce all standard diagnostic plots at once:

```
plot(model)
```

We will explore these plots in detail next week. For now, the Shapiro-Wilk and Bartlett's tests are sufficient.



## T: The ANOVA idea

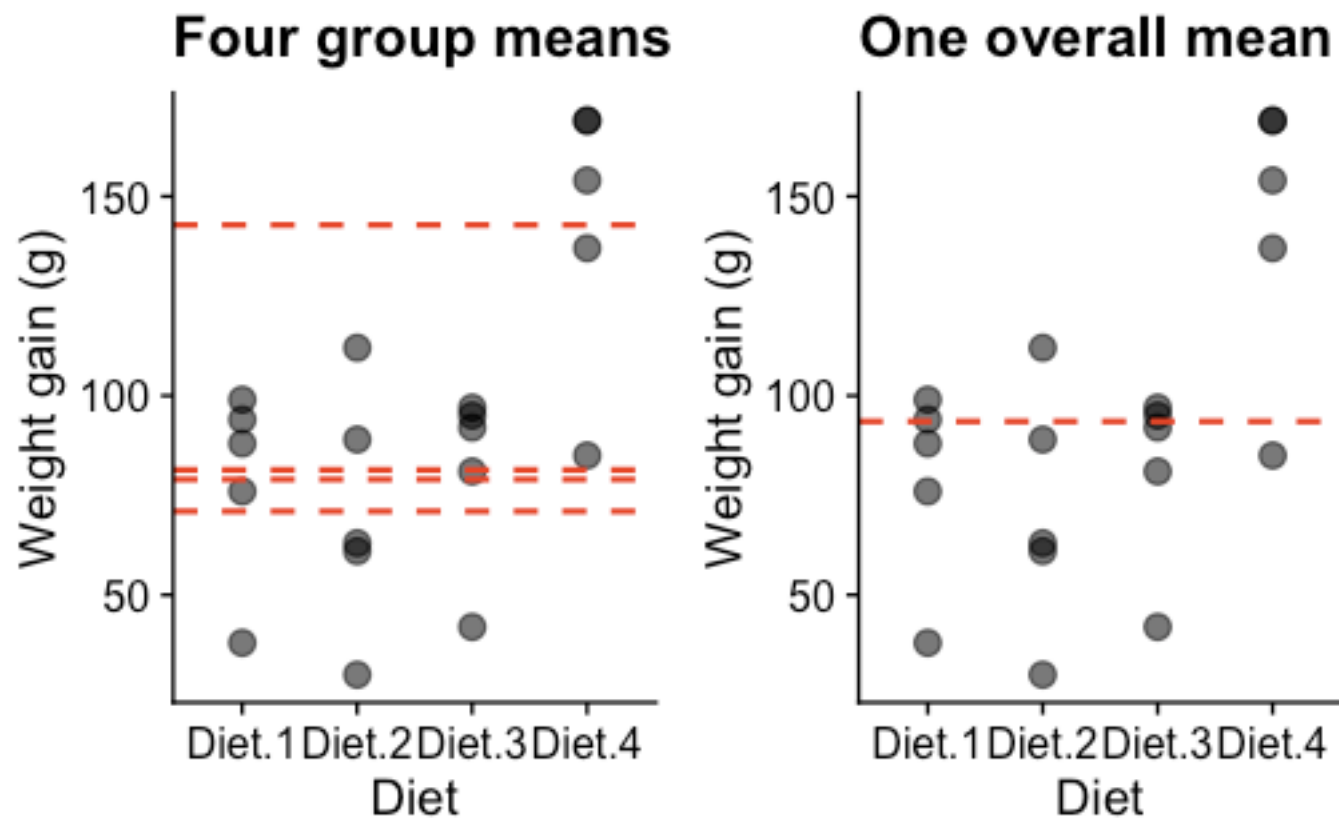
- How does ANOVA decide whether these four diets differ?
- If diets matter, differences **between** groups should be large relative to variation **within** groups

```
overall_mean ← mean(chicks$weight)
group_means ← chicks |>
  group_by(diet) |>
  summarise(mean_wt = mean(weight))

p1 ← ggplot(chicks, aes(diet, weight)) +
  geom_point(size = 3, alpha = 0.6) +
  geom_hline(data = group_means, aes(yintercept = mean_wt),
    linetype = "dashed", colour = "#e64626", linewidth = 0.8) +
  labs(title = "Four group means", x = "Diet", y = "Weight gain (g)") +
  cowplot::theme_cowplot()

p2 ← ggplot(chicks, aes(diet, weight)) +
  geom_point(size = 3, alpha = 0.6) +
  geom_hline(yintercept = overall_mean, colour = "#e64626",
    linewidth = 0.8, linetype = "dashed") +
```

```
labs(title = "One overall mean", x = "Diet", y = "Weight gain (g)") +  
cowplot::theme_cowplot()  
  
library(patchwork)  
p1 + p2
```



Do four separate means (left) explain the data better than a single overall mean (right)?

## Two sources of variation

ANOVA splits the total variation in the data into two parts:

- **Between-group variation** (orange): how far each group mean is from the overall mean. If diets matter, this should be large.
- **Within-group variation** (black): how much individual observations scatter around their own group mean. This is random noise.

ANOVA compares these two quantities. If between-group variation is large relative to within-group variation, we have evidence that the groups genuinely differ.

## Variance partitioning

```
{ojs}  
//| echo: false  
import { variancePartitionWidget } from "../assets/js/anova-widget.js"  
variancePartitionWidget()
```

## The F ratio

$$F = \frac{MS_{\text{trt}}}{MS_{\text{res}}}$$

- $F$  compares variation due to group differences against random noise
- MS (mean square) is based on **squared** deviations: squaring prevents positive and negative differences from cancelling out
- When  $H_0$  is true (all groups equal),  $F \approx 1$
- When groups genuinely differ,  $F \gg 1$

For two groups, ANOVA gives the same answer as the  $t$ -test:

$$F = t^2$$

ANOVA extends the  $t$ -test to any number of groups.

## T: The ANOVA table

| Source       | df      | SS                  | MS                          | F                                   |
|--------------|---------|---------------------|-----------------------------|-------------------------------------|
| Treatment    | $t - 1$ | $SS_{\text{trt}}$   | $SS_{\text{trt}} / (t - 1)$ | $MS_{\text{trt}} / MS_{\text{res}}$ |
| Residual     | $N - t$ | $SS_{\text{res}}$   | $SS_{\text{res}} / (N - t)$ |                                     |
| <b>Total</b> | $N - 1$ | $SS_{\text{total}}$ |                             |                                     |

```
model ← aov(weight ~ diet, data = chicks)
summary(model)
```

```

      Df Sum Sq Mean Sq F value Pr(>F)
diet      3   16467     5489   6.647  0.004 **
Residuals 16   13212      826
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## P: Interpret

- **diet** row: variation explained by the four diets ( $df = 3$ , since 4 groups minus 1)
- **Residuals** row: leftover variation ( $df = 16$ , since 20 observations minus 4 groups)
- $F = 6.65$ : the treatment variance is 6.65 times larger than the residual variance
- $p = 0.004$ : below 0.05, so we **reject**  $H_0$

## How much variability is explained?

$$\frac{SS_{\text{treatment}}}{SS_{\text{total}}} = \frac{1.6467 \times 10^4}{2.9679 \times 10^4} = 55.5\%$$

The diets explain about 55% of the total variability in chick weight gain.



## C: Conclusion

| Source   | df | SS                  | MS    | F    | p     |
|----------|----|---------------------|-------|------|-------|
| Diet     | 3  | 1.6467 <sup>4</sup> | 5489  | 6.65 | 0.004 |
| Residual | 16 | 1.3212 <sup>4</sup> | 825.8 |      |       |

There are significant differences in mean weight gain among the four diets ( $F_{3,16} = 6.65, p = 0.004$ ).

But ANOVA only tells us **that** groups differ, not **which** ones. To identify specific differences, we need post-hoc analysis.

**Post-hoc**

## Why post-hoc?

$H_1$  says “not all means are equal,” but not **which** ones differ.

- ANOVA tells us the four diet means are not all the same
- It does not tell us whether Diet 1 differs from Diet 2, or Diet 3 from Diet 4
- We cannot run separate  $t$ -tests (the multiple comparisons problem from the first slide)
- We need a method that compares pairs **while controlling the overall error rate**

## Post-hoc and confidence intervals

Before the break we used a **confidence interval for the difference** between two cattle breeds. Post-hoc methods apply the same logic to every pair of groups.

- For each pair, construct a CI for the difference in means
- If the CI **excludes zero**, those two groups differ significantly
- If the CI **includes zero**, we cannot conclude they differ

The key difference from running separate  $t$ -tests: post-hoc methods **widen** the CIs to account for multiple comparisons, keeping the overall false positive rate at 5%.

We will cover specific post-hoc methods (Tukey, Bonferroni) in detail next week.

## Estimated marginal means

Before the break we used CIs to compare two cattle breeds. The same idea extends to multiple groups. `emmeans` estimates the mean and standard error for each group from the model.

```
library(emmeans)
emm ← emmeans(model, "diet")
emm
```

| diet   | emmean | SE   | df | lower.CL | upper.CL |
|--------|--------|------|----|----------|----------|
| Diet.1 | 79.0   | 12.9 | 16 | 51.8     | 106.2    |
| Diet.2 | 71.0   | 12.9 | 16 | 43.8     | 98.2     |
| Diet.3 | 81.4   | 12.9 | 16 | 54.2     | 108.6    |
| Diet.4 | 142.8  | 12.9 | 16 | 115.6    | 170.0    |

Confidence level used: 0.95

## Pairwise comparisons

To find which pairs differ, we construct a CI for the difference between every pair of means.

```
confint(pairs(emm))
```

| contrast        | estimate | SE   | df | lower.CL | upper.CL |
|-----------------|----------|------|----|----------|----------|
| Diet.1 - Diet.2 | 8.0      | 18.2 | 16 | -44.0    | 60.0     |
| Diet.1 - Diet.3 | -2.4     | 18.2 | 16 | -54.4    | 49.6     |
| Diet.1 - Diet.4 | -63.8    | 18.2 | 16 | -115.8   | -11.8    |
| Diet.2 - Diet.3 | -10.4    | 18.2 | 16 | -62.4    | 41.6     |
| Diet.2 - Diet.4 | -71.8    | 18.2 | 16 | -123.8   | -19.8    |
| Diet.3 - Diet.4 | -61.4    | 18.2 | 16 | -113.4   | -9.4     |

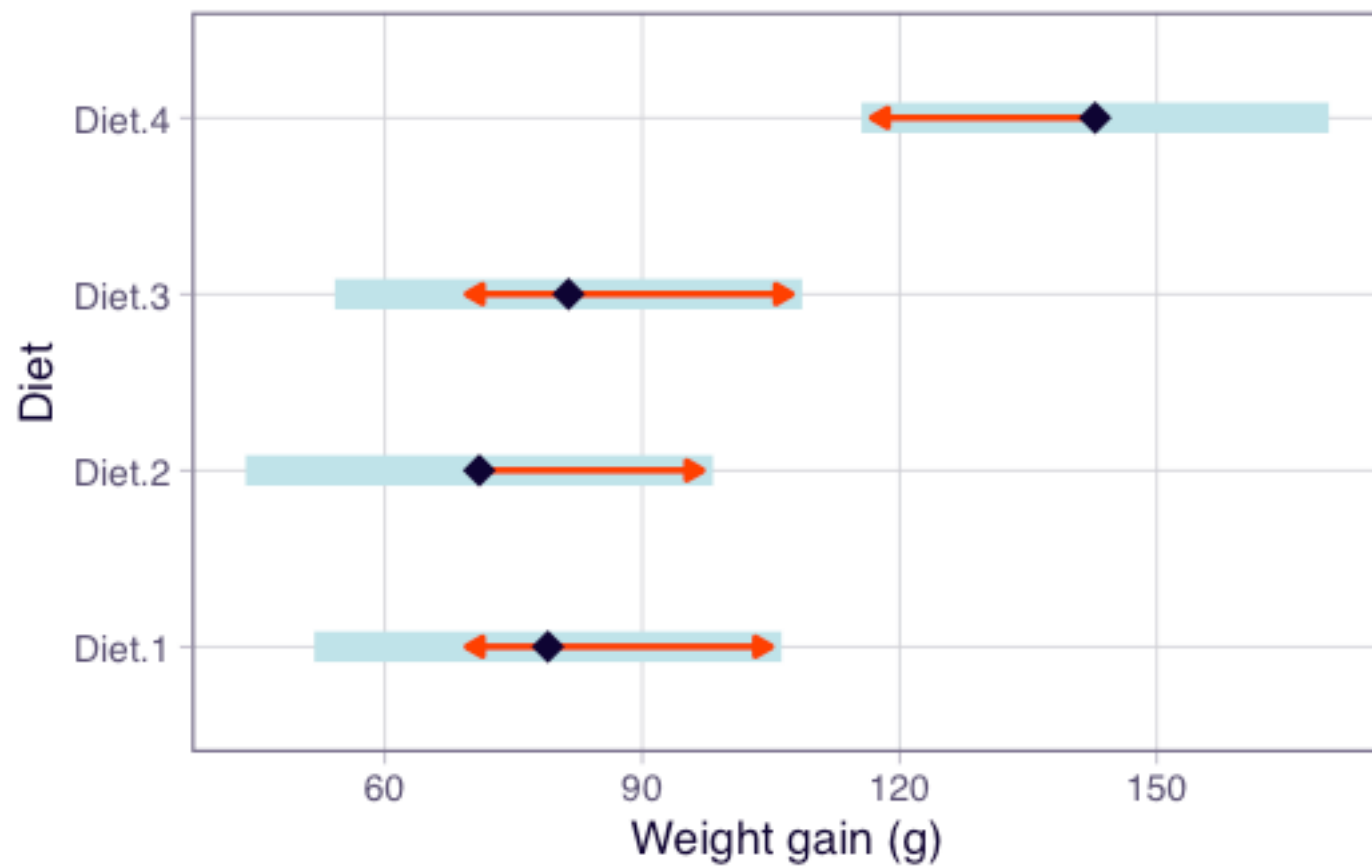
Confidence level used: 0.95

Conf-level adjustment: tukey method for comparing a family of 4 estimates

- Each row is one pairwise comparison (e.g. Diet1 - Diet2)
- The `lower.CL` and `upper.CL` columns are the 95% CI for the difference
- If the CI **excludes zero**, those two diets differ significantly

## Visualising the comparisons

```
plot(emm, comparisons = TRUE) +  
  labs(x = "Weight gain (g)", y = "Diet")
```



- Each point is the estimated mean for that diet
- The arrows represent comparison intervals (adjusted CIs)



- Groups with **overlapping** arrows are **not** significantly different
- We will cover post-hoc methods in more detail next week

# Summary

## Key points

- ANOVA extends the  $t$ -test to more than two groups ( $F = t^2$  for two groups)
- Multiple pairwise  $t$ -tests inflate the false positive rate. ANOVA tests all groups at once
- The F ratio compares between-group variation to within-group variation
- Post-hoc methods construct CIs for pairwise differences to identify which groups differ

In Lab 03, you will fit your own ANOVA with `aov()` and explore group comparisons with `emmeans`. Next week: residual diagnostics and post-hoc methods.

**Thanks!**

**Questions?**

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